

Experimental apparatus for universal spin-motion control in multi-ion registers



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Theory

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1.1 Trapping Ions

1.1.1 Trapping Potential

From Earnshaw's theorem, $\nabla^2 V = 0$, a stable stationary point in 3D cannot be realised using only static electric potentials, V , as if the potential is confining in two dimensions, it will be anti-confining in the third. Therefore, to achieve stable trapping, either an oscillating electric field (Paul trap [paul1990electromagnetic]), or a static magnetic field (Penning trap) is used.

Will include pseudopotential relating V_{RF} and Ω_{RF} to the radial mode frequencies ω_m (see Vera eqn 3.4),

$$\omega_r \propto \frac{V_{RF}}{M\Omega_{RF}}. \quad (1.1)$$

Further, will use James to give ion spacing and non-COM mode frequencies as will be relevant for single addressing, fast gates (if discussed), and for motional mode excitati as will be relevant for single addressing, fast gates (if discussed), and for motional mode excitation.

1.2 Light Matter Interactions

1.2.1 Rabi Flopping

1.2.2 Motional Coupling

1.2.3 Geometric Phase Gates

1.3 Randomised Benchmarking

Some overview and motivation for RB. Need to cover:

- Twirling over a group
- Liouville representation of a CPTP map

- Irreps of the Clifford group
- Clifford approximates twirling over the unitary group (Haar measure)
- How Liouville representation is transformed under twirling
- The specific representation-theory tools you use later: irreps, multiplicity spaces, invariant sectors, Schur basis.
- RB protocol for single qubit Clifford group, generalisation to multi-qubit Clifford group.
- How to extract error per Clifford from decay rate

1.4 Noise channels

As will also be used in virtual distillation, may move to theory sections

A description of the noise channels we consider in this work. Explicit CPTP maps for these channels.

1.4.1 Why isn't everything white noise?

1.4.2 Unitary noise

Both continuous and probabilistic.

1.4.3 Depolarising noise

An ensemble of unitary noise channels, with a uniform distribution over the Bloch sphere. Get this from twirling over Clifford group!

1.4.4 Spatially correlated noise

Common noise An ensemble of unitary noise channels, with a uniform distribution over the Bloch sphere, but the same unitary is applied to multiple qubits. Can enforce some symmetry to this channel by twirling over the k-copy Clifford group.

This is the channel we will use to model correlated noise in our experiments.

Entangling noise In SC ZZ entangling noise is common.

1.4.5 Can we make everything white noise?

Have a look at Balint paper, deep random circuits, etc.

1.5 v1

This chapter lays the theoretical groundwork for Chapter ?? . It reviews standard Clifford randomised benchmarking (RB), develops the representation-theoretic machinery of group twirling in some generality before specialising it to the two-copy Clifford group, and surveys the classes of noise channel and prior characterisation strategies that motivate a symmetry-based approach to correlated-noise detection.

1.6 Standard Clifford randomised benchmarking

1.6.1 Motivation: characterising noise without full tomography

As quantum processors scale, fault tolerance and error mitigation become necessary to overcome decoherence and systematic control errors [1–3]. Errors that induce correlations across multiple qubits — variously termed spatial correlations or crosstalk errors [4] — require that mitigation and correction strategies be built on noise models that faithfully capture their correlated structure. Assuming a globally depolarising (white) noise model where correlations are in fact present can lead to systematic underestimation of logical error rates [5], or systematic bias in the outcome of an error-corrected observable [6].

Methods exist to simplify correlated noise channels into more favourable depolarising or white-noise forms, for example Pauli twirling and random circuit compilation [7–9]. However, the efficacy of such methods must still be characterised, verified, and optimised on physical devices with realistic noise, which is only possible with a scalable diagnostic for the correlated component of that noise in the first place.

Add a short paragraph situating this thesis chapter within the broader RB literature landscape (interleaved RB, cycle benchmarking, purity benchmarking, etc.), with a comparison table of scaling/expressiveness trade-offs. This context was compressed to a single sentence in the PRL and would benefit from a fuller literature review here.

1.6.2 The single-qubit Clifford RB protocol

Standard single-qubit Clifford RB proceeds by applying a sequence of $M - 1$ Cliffords, sampled uniformly at random from the single-qubit Clifford group, followed by a single inverting Clifford, C_{inv} , chosen so that the sequence implements the identity operation in the absence of errors [10, 11]. For each sequence length M , r independent random sequences are generated, and each is repeated s times to estimate the probability of the qubit returning to its target computational state. The target state is conventionally randomised between $|0\rangle$ and $|1\rangle$ from sequence to sequence, which reduces sensitivity to asymmetries in state preparation and measurement [12].

Circuit diagram: single-qubit Clifford RB sequence of length M , with per-step error box and final inverting Clifford C_{inv} . Schematic single-qubit Clifford RB circuit (reproduced/adapted from Fig. 1(a) of the manuscript). **Insert final figure once redrawn for thesis formatting; the manuscript version also shows k parallel copies, which belongs in Chapter ?? instead.**

The survival probability of a single qubit after m Cliffords follows a single exponential decay,

$$\mathcal{P}_0 = \frac{1}{2} (1 + (1 - \varepsilon_c)^m), \quad (1.2)$$

where ε_c is the “error per Clifford” (EPC), the single scalar figure of merit that standard RB is able to extract. This compression of an entire error channel into one averaged parameter is precisely what gives RB its scalability, but it is also the limitation that motivates the correlated-noise-resolving protocol developed in Chapter ??: two qubits with identical single-qubit EPC can have entirely different amounts of correlated error between them, and Eq. (1.2) cannot distinguish these cases.

1.6.3 State-preparation and measurement (SPAM) errors

RB's scalability partly derives from its insensitivity to SPAM errors up to an overall scaling of the exponential decay. Modelling independent bit-flip type state-preparation error on a single qubit with probability $\varepsilon_{sp}^{(i)}/2$, the prepared state is

$$\rho_{sp}^{(i)} = \left(1 - \frac{\varepsilon_{sp}^{(i)}}{2}\right) |0\rangle\langle 0| + \frac{\varepsilon_{sp}^{(i)}}{2} |1\rangle\langle 1|, \quad (1.3)$$

equivalently expressed as a probabilistic X channel acting on an arbitrary input state,

$$\rho \mapsto \mathcal{E}_{sp}^{(i)}(\rho) = \left(1 - \frac{\varepsilon_{sp}^{(i)}}{2}\right) \rho + \frac{\varepsilon_{sp}^{(i)}}{2} X\rho X. \quad (1.4)$$

In the Liouville picture this simply rescales the overlap of the initial state with each sub-representation of the twirled channel by a factor $A_{sp}^{(i)} = (1 - \varepsilon_{sp}^{(i)})$ for every sub-representation in which qubit i participates non-trivially. This single-qubit treatment generalises directly to the two-qubit setting used in Chapter ??: correlated SPAM errors, such as a correlated spin-flip channel, leave certain sub-representations (in particular the fully symmetric sector) invariant, and so can mimic a genuine correlated error signature if left unaccounted for in a fit.

Expand this subsection with the general n -qubit SPAM formalism, and a short discussion of readout-error mitigation strategies compared against this multiplicative-scaling picture. The manuscript SM only develops the 2-qubit special case, which is deferred to Chapter ??.

1.7 Twirling and representation theory

1.7.1 General twirling formalism

RB owes its practicality to the process of *twirling*: averaging a general noise channel over a chosen group of unitaries. Consider a general CPTP error channel Λ , expressed in Liouville (transfer-matrix) form acting on a vectorised density matrix. Twirling Λ over a discrete group $\mathcal{G}$ produces

$$\Lambda \mapsto \tilde{\Lambda} = \frac{1}{|\mathcal{G}|} \sum_{\mathcal{G}} G\Lambda G^\dagger, \quad (1.5)$$

where G denotes the vectorised representation of a group element. By construction, the twirled channel commutes with every representation element,

$$G\tilde{\Lambda}G^\dagger = \tilde{\Lambda} \quad \forall G \in \mathcal{G}. \quad (1.6)$$

The Hilbert space on which $\mathcal{G}$ acts decomposes into a direct sum of irreducible representations (irreps),

$$\mathcal{H} = \bigoplus_{\lambda} (V_{\lambda} \otimes \mathbb{C}^{m_{\lambda}}), \quad (1.7)$$

where V_{λ} is the irrep space and $\mathbb{C}^{m_{\lambda}}$ is a multiplicity space of dimension m_{λ} . Schur's lemma then constrains any operator satisfying Eq. (1.6) to be block diagonal in this decomposition,

$$\tilde{\Lambda} = \bigoplus_{\lambda} (I_{\lambda} \otimes \mathcal{A}_{\lambda}), \quad (1.8)$$

where I_{λ} is the identity on V_{λ} and $\mathcal{A}_{\lambda}$ is an $m_{\lambda} \times m_{\lambda}$ matrix acting only within the multiplicity space. Physically, Eq. (1.8) is the statement that twirling collapses an otherwise arbitrarily complex error channel down to a comparatively small number of scalar decay rates (diagonal terms of $\mathcal{A}_{\lambda}$) and couplings (off-diagonal terms), one set per irrep. This is the general mechanism underlying the scalability of every RB variant, and the choice of twirling group $\mathcal{G}$ determines which physical information survives the averaging and which is discarded.

Add a short worked example here with a simple, well-known group (e.g. the single-qubit Clifford group acting on one qubit, recovering standard RB's single decay parameter) to build intuition before jumping to the two-copy case below. This would also be a natural place to discuss the trade-off between expressiveness and scalability more formally, referencing Pauli-twirling as an alternative choice of $\mathcal{G}$.

1.7.2 The two-copy Clifford group and its irreducible representations

To characterise correlated noise between a pair of qubits, we are interested in the group generated by applying the *same* randomly sampled single-qubit Clifford

Sub-representation	irrep	Basis operators
Identity	1	II
Left qubit	3	XI, YI, ZI
Right qubit	3	IX, IY, IZ
Trace	1	$(XX + YY + ZZ)/\sqrt{3}$
Diagonal traceless	2	$(XX + YY - 2ZZ)/\sqrt{6}, (XX - YY)/\sqrt{2}$
Symmetric off-diagonal	3'	$(XY + YX)/\sqrt{2}, (XZ + ZX)/\sqrt{2}, (YZ + ZY)/\sqrt{2}$
Antisymmetric	3	$(XY - YX)/\sqrt{2}, (XZ - ZX)/\sqrt{2}, (YZ - ZY)/\sqrt{2}$

Table 1.1: Sub-representations of the $Cl^{\otimes 2}$ group, adapted from Eq. 56 of Ref. [14]. Each row gives the irrep label, its dimension, and a spanning set of two-qubit Pauli-basis operators.

irrep	e	(12)	(12)(34)	(123)	(1234)
Identity/Trace	1	1	1	1	1
Left/Right/Antisymmetric	3	0	-1	1	1
Diagonal traceless	2	-1	2	0	0
Symmetric off-diagonal	3	0	-1	-1	-1

Table 1.2: Character table for the irreps of Table 1.1, listed against representative conjugacy classes: identity, transposition, double transposition, 3-cycle, and 4-cycle.

simultaneously to both qubits: the diagonal action of the Clifford group on two copies, denoted $Cl^{\otimes 2}$. The representation theory of this group has been analysed in detail elsewhere [13, 14]; the key results are summarised here.

The general 16-dimensional two-qubit Liouville space decomposes under $Cl^{\otimes 2}$ into four irreducible representations, of dimensions $\{1, 2, 3, 3'\}$ (the label $3'$ distinguishes the second, inequivalent, three-dimensional irrep from the first), with respective multiplicities $\{2, 1, 3, 1\}$. Table 1.1 lists the six sub-representations that follow from these irreps and multiplicities, together with a spanning set of Pauli-basis operators for each.

Insert the derivation connecting Tables 1.1 and 1.2 explicitly (i.e. show the character-table computation step by step, rather than quoting the result). The manuscript SM states this result without full derivation; a thesis chapter is the appropriate place to walk through it, likely via Young-tableaux / permutation-representation arguments.

Given the block-diagonal structure of Eq. (1.8), and further imposing that the twirled channel be trace preserving, a general twirled two-qubit channel is described

Coupling	irrep	Mechanism
$m_{id \rightarrow tr}$	1	Decays (non-unital)
$m_{L \rightarrow R}, m_{R \rightarrow L}$	3	Qubit swap, correlated decays (non-unital)
$m_{L \rightarrow as}, m_{R \rightarrow as}, m_{as \rightarrow L}, m_{as \rightarrow R}$	3	Entanglement

Table 1.3: Physical mechanisms capable of coupling sub-representation sectors of the twirled $Cl^{\otimes 2}$ error channel.

by 13 independent scalars: six diagonal decay rates, $\vec{\alpha} = \{\alpha_L, \alpha_R, \alpha_{tr}, \alpha_{dt}, \alpha_{so}, \alpha_{as}\}$, one per sub-representation, and seven off-diagonal coupling terms, $\vec{m} = \{m_{id \rightarrow tr}, m_{L \rightarrow R}, m_{R \rightarrow L}, m_{L \rightarrow as}, m_{R \rightarrow as}, m_{as \rightarrow L}, m_{as \rightarrow R}\}$, describing transitions between multiplicity spaces of the same irrep. Table 1.3 associates each coupling term with a physical error mechanism capable of generating it.

This decomposition is a purely group-theoretic statement: it is agnostic to any particular physical error model and simply enumerates the possible subspaces that a two-qubit error channel can populate once diagonal Clifford twirling has been applied. Section 1.8 below specialises this general structure to specific, physically motivated error channels, and Chapter ?? uses the resulting scalar decay rates to build the observable population dynamics measured experimentally.

Add a subsection discussing the multiplicity structure in more depth — i.e. what it physically means for an irrep to appear with multiplicity > 1 , and how this connects to the existence of the coupling terms $\vec{m}$. Also worth adding a small worked example reducing a simple noise channel (e.g. pure local depolarisation) through Eq. (1.8) by hand for pedagogical value.

1.8 Noise models

Having established the general block-diagonal structure available to any twirled two-qubit channel, this section reviews the specific classes of physical error mechanism relevant to the results in Chapter ??, and how each maps onto the sub-representations of Table 1.1.

1.8.1 Local depolarising noise

Independent noise acting separately on each qubit is, without loss of generality (following twirling), well modelled by a depolarising channel,

$$\rho^{(1q)} \mapsto \mathcal{E}_{dp,a}(\rho^{(1q)}) = (1 - \varepsilon_a)\rho^{(1q)} + \varepsilon_a \frac{I}{2} \text{Tr}(\rho^{(1q)}), \quad (1.9)$$

where the subscript $a \in \{L, R\}$ labels the qubit and ε_a is its depolarising error per Clifford. In Liouville form, restricted to the two-qubit Pauli basis $\mathcal{P} = \{I \otimes I, \vec{\sigma} \otimes I, I \otimes \vec{\sigma}, \vec{\sigma} \otimes \vec{\sigma}\}$, independent depolarisation on both qubits gives

$$\Lambda_{dp} = \text{diag}^{\mathcal{P}} \left(1, (1 - \varepsilon_L)I_3, (1 - \varepsilon_R)I_3, (1 - \varepsilon_L)(1 - \varepsilon_R)I_9 \right). \quad (1.10)$$

Local depolarising noise generates no couplings between sub-representations: it is diagonal in the Schur basis introduced in Section 1.7.2, and drives the two-qubit state toward the fully mixed state.

1.8.2 Correlated (“common”) single-qubit rotation noise

The central noise model of interest is correlated random-unitary noise, in which the *same* randomly sampled single-qubit rotation error is applied to both qubits, here termed *common noise*. This channel is

$$\rho \mapsto \mathcal{E}_{U \otimes U}^{\mathcal{A}}(\rho) = \sum_{a \in \mathcal{A}} p_a (U_a \otimes U_a) \rho (U_a \otimes U_a)^\dagger, \quad (1.11)$$

where $\mathcal{A}$ is an ensemble of unitaries applied with probabilities p_a ($\sum_a p_a = 1$), and each $U_a(\theta_a, \hat{n}_a) = e^{-i(\theta_a/2)\hat{n}_a \cdot \hat{\sigma}}$ is a rotation of angle θ_a about axis $\hat{n}_a$. A subset $\mathcal{A}' \subseteq \mathcal{A}$ of this class produces genuinely non-separable two-qubit channels, i.e. $\mathcal{E}_{U \otimes U}^{\mathcal{A}'} \neq \mathcal{E}_U^{(L)} \otimes \mathcal{E}_U^{(R)}$, and it is precisely this subset that constitutes spatially correlated noise.

Physically, this class captures correlated but non-entangling control and environmental fluctuations, which are particularly relevant to trapped-ion and neutral-atom qubits subject to common fluctuations of qubit drive phase, amplitude, and detuning; magnetic field noise and gradients; beam pointing and intensity drift; AC Stark-shift fluctuations; motional-state-dependent over-rotations; and gate-duration calibration errors.

Twirling Eq. (1.11) over $Cl^{\otimes 2}$ reveals that only the fully symmetric “trace” sub-representation is exactly preserved by common single-qubit rotations, $\alpha_{tr} = 1$, regardless of the error angle. To second order in a small rotation angle θ , the remaining scalar decay rates for a single common rotation are

$$\alpha_L = \alpha_R = \alpha_{as} = 1 - \theta^2/3, \quad \alpha_{tr} = 1, \quad \alpha_{dt} = \alpha_{so} = 1 - \theta^2, \quad (1.12)$$

so that, to leading order, the dynamics depend only on the magnitude of the error angle and not on its axis. Averaging over the ensemble $\mathcal{A}$ gives $\alpha_i^{(\mathcal{A})} = \sum_a p_a \alpha_a$, leading to the convenient definition of a correlated error per Clifford,

$$\varepsilon_{\text{corr}} = \frac{1}{3} \sum_{a \in \mathcal{A}} p_a \theta_a^2. \quad (1.13)$$

Add the full (non-small-angle) expressions here for completeness, i.e. $\alpha_L = \alpha_R = \alpha_{as} = (1 + 2 \cos \theta)/3$ and the corresponding α_{dt} , α_{so} including the anisotropic term n_{an} , together with a short discussion of when the small-angle approximation breaks down. Also worth adding the tensor-contraction derivation ($s_{tr}, s_{id}, s_{swap}, s_{diag}$) that produces these α_i from first principles, since it is a reusable technique beyond this specific channel.

1.8.3 Entangling coherent errors: ZZ-type noise

A distinct and experimentally important source of two-qubit correlation is coherent entangling coupling, exemplified in superconducting-qubit architectures by always-on ZZ crosstalk [fors_comprehensive_2024, alghadeer_crosstalk_2025, 15], described by the Hamiltonian

$$H_{ZZ} = \frac{J_{ZZ}}{2} Z \otimes Z. \quad (1.14)$$

Unlike common single-qubit rotation noise, H_{ZZ} couples the two multiplicity copies of the 3-dimensional irrep (the “left”, “right”, and “antisymmetric” sub-representations of Table 1.1), and so cannot in general be described by a diagonal Liouville matrix in the Schur basis. A more tractable, stochastic variant of this error,

$$\rho \mapsto \mathcal{E}_{ZZ}(\rho) = (1 - p_{ZZ})\rho + p_{ZZ} (Z \otimes Z)\rho(Z \otimes Z), \quad (1.15)$$

does remain diagonal, with

$$\alpha_L = \alpha_R = \alpha_{so} = \alpha_{as} = 1 - 4p_{ZZ}/3, \quad \alpha_{tr} = \alpha_{dt} = 1. \quad (1.16)$$

This channel is included here for completeness and generality; it is not probed experimentally in this thesis (see Chapter ??, Outlook).

Expand this subsection substantially if ZZ -error characterisation numerics or a proposed experiment are added elsewhere in the thesis. As it stands this is a very compressed treatment inherited directly from the manuscript SM; a thesis-length treatment might derive the full coherent (non-stochastic) block-diagonal Liouville matrix, rather than only the stochastic special case.

2

k-copy Randomised Benchmarking

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Building on the general twirling formalism and noise models of Chapter ??, this chapter introduces *k-copy randomised benchmarking*: an RB protocol in which identical single-qubit Clifford sequences are applied in parallel across multiple qubits, and pairwise measurement correlations are retained rather than averaged away. We derive the resulting population dynamics for the common-noise model of Section 1.8.2, and present experimental demonstrations on a trapped-ion processor, including isolation of correlated versus local error, a parity-based confirmation of

Parameter	Value
729-nm laser power	1000 μ W

Table 2.1: Summary table of other methods

the underlying symmetric-subspace occupation, a controlled noise-injection scan, and scaling to an eight-ion register.

2.1 Existing approaches to correlated-noise characterisation

Define spatial correlations (do in theory section?)

Existing approaches to correlated-noise characterisation span a trade-off between expressiveness and scalability. Tomographic and noise-learning techniques can reconstruct detailed structure in the error channel, but their cost grows rapidly with system size [16–19]. Random circuit methods, including randomised benchmarking, are substantially more scalable, but typically compress the action of noise into a small number of averaged parameters [4, 11, 20–22].

A related RB protocol, termed “Correlated RB” [20] (itself built on “Simultaneous RB” [22]), uses *independent* local twirling of multiple system copies to estimate the distance of the measured channel from a product (uncorrelated) channel. This differs fundamentally from the *identical*-sequence twirling strategy developed in Chapter ??: independent local twirling averages away exactly the symmetry information that identical twirling preserves.

A complementary source of inspiration comes from correlation spectroscopy and decoherence-free-subspace techniques, where common-mode noise is revealed through its action on collective and differential degrees of freedom [23]. In such settings, symmetry is not merely a simplifying assumption: it is the very mechanism that separates common fluctuations from local noise. This observation — that enforcing a symmetry can make correlated dynamics directly visible without reconstructing the full many-qubit channel — is the guiding idea carried into Chapter ??, where it is realised concretely as *k*-copy randomised benchmarking.

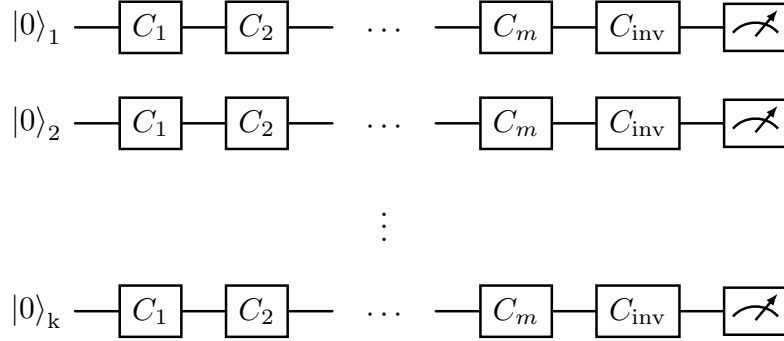


Figure 2.1: Circuit diagram for *k*-copy RB.

Add a comparative summary table at the end of this section (methods $\times$ {scalability, expressiveness, assumptions, experimental requirements}) contrasting tomography, Simultaneous RB, Correlated RB, cycle benchmarking, and *k*-copy RB. This would give the reader a single reference point before moving into Chapter ??.

2.2 The *k*-copy RB protocol

We introduce a symmetry-based viewpoint into randomised benchmarking through *k*-copy RB. In *k*-copy RB, the same single-qubit Clifford sequence is applied in parallel to multiple qubits. This diagonal Clifford twirl preserves symmetry sectors that are invisible to standard single-qubit RB (Section 1.6) but become accessible through pairwise measurement outcomes. For two qubits, this symmetry separates local single-qubit decay from collective two-copy dynamics.

Concretely, the protocol applies a standard single-qubit Clifford RB sequence [10, 11] simultaneously to multiple qubits, with the *same* randomly sampled Clifford applied to every qubit at each step. For each sequence length M , r random Clifford sequences are generated and each appended with its inverting Clifford, as in conventional RB, with each sequence repeated s times. As in standard RB,

the target computational state is randomised between $|0\rangle$ and $|1\rangle$ per sequence to reduce sensitivity to SPAM asymmetries (Section 1.6.3).

(a) Circuit diagram for k -qubit parallel Clifford RB. (b) Schematic single-qubit final states before measurement, sequence by sequence. (c) Bloch-sphere ensembles for correlated vs. uncorrelated qubit pairs. The k -copy protocol: identical parallel single-qubit Clifford RB sequences applied across k qubits (adapted from manuscript Fig. 1). **Redraw for thesis; consider splitting panel (c) out into its own figure with expanded discussion of the joint-angle-distribution picture (see note at end of Section 2.2.1).**

The distinguishing feature of k -copy RB relative to standard or Simultaneous RB (Section 2.1) is the combination of identical Clifford sequences with shot-resolved post-processing of *correlated* outcomes. For each qubit pair (i, j) , every shot is assigned to one of four outcomes, $P_{SS}^{(i,j)}$, $P_{SF}^{(i,j)}$, $P_{FS}^{(i,j)}$, or $P_{FF}^{(i,j)}$, where S and F denote success and failure relative to the sequence-dependent target outcome. Standard single-qubit RB averages this data into independent survival probabilities (Eq. (1.2)), collapsing the dynamics to a single exponential decay. By retaining pairwise correlations between successes and failures instead, k -copy RB distinguishes shared, sequence-dependent deviations from independent local decay. For a register of k qubits, a single shot record simultaneously contains outcomes for all $k(k-1)/2$ pairs, so extracting pairwise correlations requires only additional classical post-processing and no extra trials on the quantum register (this scaling property is used explicitly in Section 2.5.3).

Formally, applying identical Cliffords across k qubits implements the twirled representation of the single-qubit Clifford group, $Cl^{\otimes k}$. For two qubits, this is exactly the representation analysed generally in Section 1.7.2: the twirl decomposes the two-qubit Liouville space into the symmetry-informed subspaces of Table 1.1. The remainder of this chapter uses that decomposition, together with the common-noise model of Section 1.8.2, to derive concrete, fittable predictions for the correlated populations measured in a k -copy RB experiment.

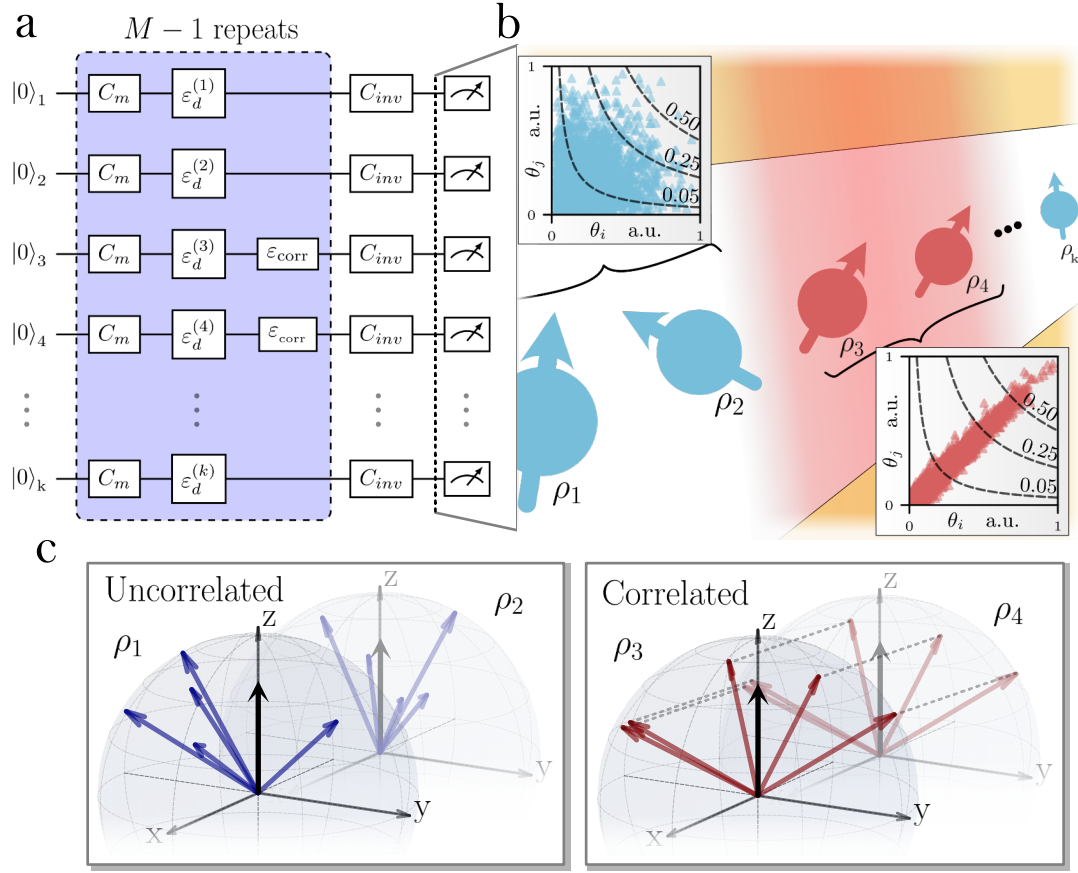


Figure 2.2: Bloch sphere schematic of corr vs uncorr noise. Include schematic of ion chain prev fig 1c.

Add a short worked numerical example here (e.g. a simulated two-qubit trace showing $P_{SS}, P_{SF}, P_{FS}, P_{FF}$ for a chosen $\{\varepsilon_L, \varepsilon_R, \varepsilon_{\text{corr}}\}$) to make the abstract shot-bookkeeping description concrete before moving to the analytic derivation.

2.2.1 Illustrating correlated vs. uncorrelated signatures

The distinction between correlated and uncorrelated noise is illustrated schematically by considering a register in which qubits $i = 3$ and $i = 4$ are dominated by a common rotation error while other qubits are dominated by independent noise. All qubits can exhibit identical single-qubit RB error rates (i.e. an identical depolarised Bloch vector under standard RB), while possessing entirely different pairwise correlation structure: independent noise produces uncorrelated sequence-to-sequence deviations,

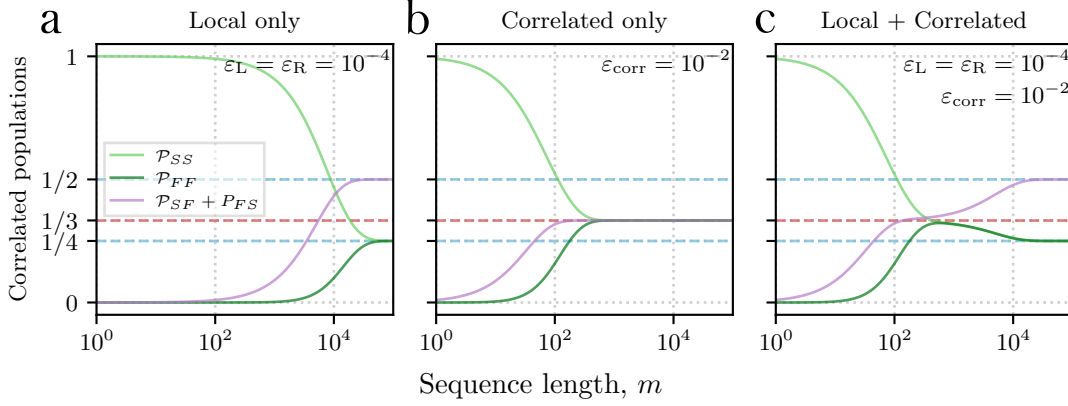


Figure 2.3: Numerical simulation of intermediate asymptotes measured in Z basis.

whereas common noise produces shared deviations between qubits acted on by the same random Clifford sequence. The identical-sequence condition is therefore essential to the protocol: it preserves exactly the symmetry information that would be averaged away under independently sampled local Clifford sequences (as in Simultaneous/Correlated RB, Section 2.1).

This is currently a qualitative, schematic argument (matching the level of detail in the PRL). A thesis chapter could usefully add the joint-angle-distribution analysis that was drafted but cut from the manuscript: uncorrelated noise exponentially suppresses higher-weight error terms $p_w = \prod_i p_i$ relative to single-qubit rates, whereas correlated noise violates this independence assumption and can scale linearly with weight w in the worst case, with consequences for QEC thresholds assuming independent errors ($p_t \propto \theta_i^2 \theta_j^2$). This connects naturally to the > 2 -qubit scaling results in Section 2.5.3.

2.3 Analytic predictions for common noise

2.3.1 Correlated population decay

Using the block-diagonal Liouville structure of Eq. (1.8) and the scalar decay rates it implies, the probability of both qubits successfully returning to their initial state

after m Cliffords, for a general error channel diagonal in the irrep basis of Table 1.1, is

$$\mathcal{P}_{SS} = \frac{1}{4} + \frac{1}{4}(\alpha_L^m + \alpha_R^m) + \frac{1}{6}\alpha_{dt}^m + \frac{1}{12}\alpha_{tr}^m, \quad (2.1)$$

with corresponding expressions for $\mathcal{P}_{SF}$, $\mathcal{P}_{FS}$, and $\mathcal{P}_{FF}$ following the same structure with sign changes on the α_L^m, α_R^m and $\alpha_{dt}^m, \alpha_{tr}^m$ terms respectively.

Combining the local depolarising channel (Eq. (1.10)) with the twirled common-noise channel (Section 1.8.2) gives, to first order, the sector decay rates

$$\alpha_L = 1 - \varepsilon_L - \varepsilon_{\text{corr}}, \quad \alpha_R = 1 - \varepsilon_R - \varepsilon_{\text{corr}}, \quad (2.2)$$

$$\alpha_{tr} = 1 - \varepsilon_L - \varepsilon_R, \quad \alpha_{dt} = 1 - \varepsilon_L - \varepsilon_R - 3\varepsilon_{\text{corr}}, \quad (2.3)$$

where $\varepsilon_{\text{corr}}$ is defined by Eq. (1.13). Because the symmetric trace sector is exactly invariant under common rotation noise ($\alpha_{tr} = 1$ when $\varepsilon_L = \varepsilon_R = 0$), Eq. (2.1) predicts an intermediate asymptote, $\mathcal{P}_{SS}|_{m \rightarrow \infty} = 1/3$, distinct from the fully depolarised asymptote $\mathcal{P}_{SS}|_{m \rightarrow \infty} = 1/4$ reached once any finite local error rate is present. This intermediate asymptote is the central experimental signature of correlated noise exploited throughout this chapter, and corresponds to the two-qubit state

$$\rho_{\text{sym}} = \frac{1}{3} \left(|00\rangle\langle 00| + |11\rangle\langle 11| + |\Psi^+\rangle\langle \Psi^+| \right), \quad (2.4)$$

where $|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$ is a Bell state. The presence of this Bell-state component reflects residual coherence induced between the two qubits by the shared rotation error, and motivates the independent, coherence-sensitive probe developed in Section 2.3.2.

Insert the full derivation connecting Eq. (2.1) to the vectorised-state inner product $\mathcal{P}_{L,R} = \langle\langle L, R | \Lambda_T^m | 00 \rangle\rangle$ and the explicit transfer matrices, rather than quoting the result (this is worked in the manuscript SM and can be reproduced here in full for thesis completeness, including the $\mathcal{P}_{0,1}, \mathcal{P}_{1,0}, \mathcal{P}_{1,1}$ companion expressions).

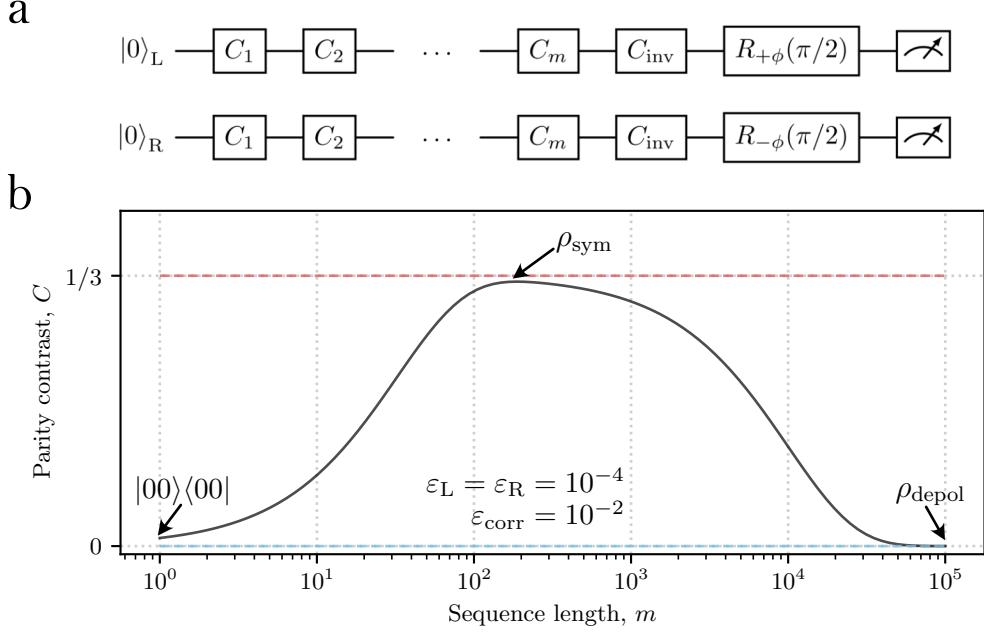


Figure 2.4: circuit diagram for parity measurement. Could there also be a nice schematic of the Bloch sphere picture of the differential rotation?

2.3.2 Parity contrast: an independent probe of coherence

Population decays alone indicate an intermediate asymptote; a parity-based measurement independently verifies that this asymptote corresponds specifically to occupation of the symmetric trace sector, by directly probing the residual coherence $|01\rangle\langle 10|$ built up between the two qubits. Differential rotations of the form $\hat{R}_{+\phi}(\pi/2) \otimes \hat{R}_{-\phi}(\pi/2)$, with rotation axis $\hat{\sigma}_\phi$ in the XY -plane, are applied prior to measurement, and the contrast of the resulting fringe (as ϕ is varied) is extracted. This contrast has the predicted dependence

$$\mathcal{C} = \frac{1}{3} (\alpha_{tr}^m - \alpha_{dt}^m), \quad (2.5)$$

with a maximum value $\mathcal{C}_{\text{max}} = 1/3$, reached in the same regime in which the population asymptote of Eq. (2.1) sits at $1/3$. Equations (2.1) and (2.5) may be combined to isolate the individual sector decay rates α_{tr} and α_{dt} as single exponentials, rather than fitting the multi-exponential population decay directly.

Reproduce the full parity-contrast derivation from the manuscript SM (App. “Parity analysis”): the transfer-matrix expression $\Pi(\phi) =$

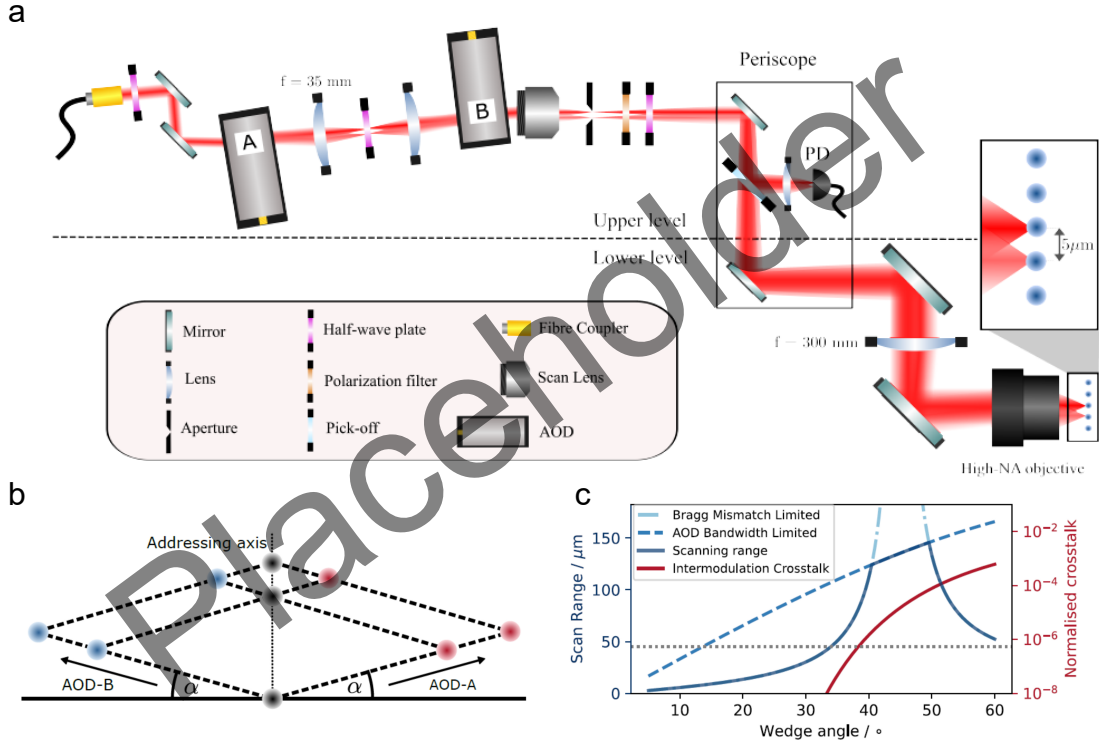


Figure 2.5: Block diagonal Liouville structure of the two-copy Clifford twirl, showing the sub-representations and their associated decay rates.

$\langle\langle\pi|\Lambda_{\Pi}(\phi)\Lambda_T^m|00\rangle\rangle = \mathcal{C} \cos(2\phi)$, the Bloch-sphere-averaging argument giving $\mathcal{C}_{\max} = 1/3$, and the isolation identities $\Pi_Z - C = \alpha_{dt}^m$, $\Pi_Z + 2C = \alpha_{tr}^m$ using the computational parity $\Pi_Z = \mathcal{P}_{0,0} + \mathcal{P}_{1,1} - \mathcal{P}_{0,1} - \mathcal{P}_{1,0}$. This is a good candidate for a fully worked thesis subsection given its role in confirming the central experimental claim.

2.3.3 Isolating sectors via initial-state choice

Beyond parity measurements, the sector decomposition of Table 1.1 suggests that preparing initial states with overlap on only a subset of sub-representations can isolate individual decay constants directly, without needing to fit multi-exponential curves. For example, preparing the singlet state $|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$ gives a density matrix with overlap only on the identity and symmetric-trace sub-representations, so that the correlated populations decay as a single exponential in α_{tr} alone, insensitive to $\varepsilon_{\text{corr}}$.

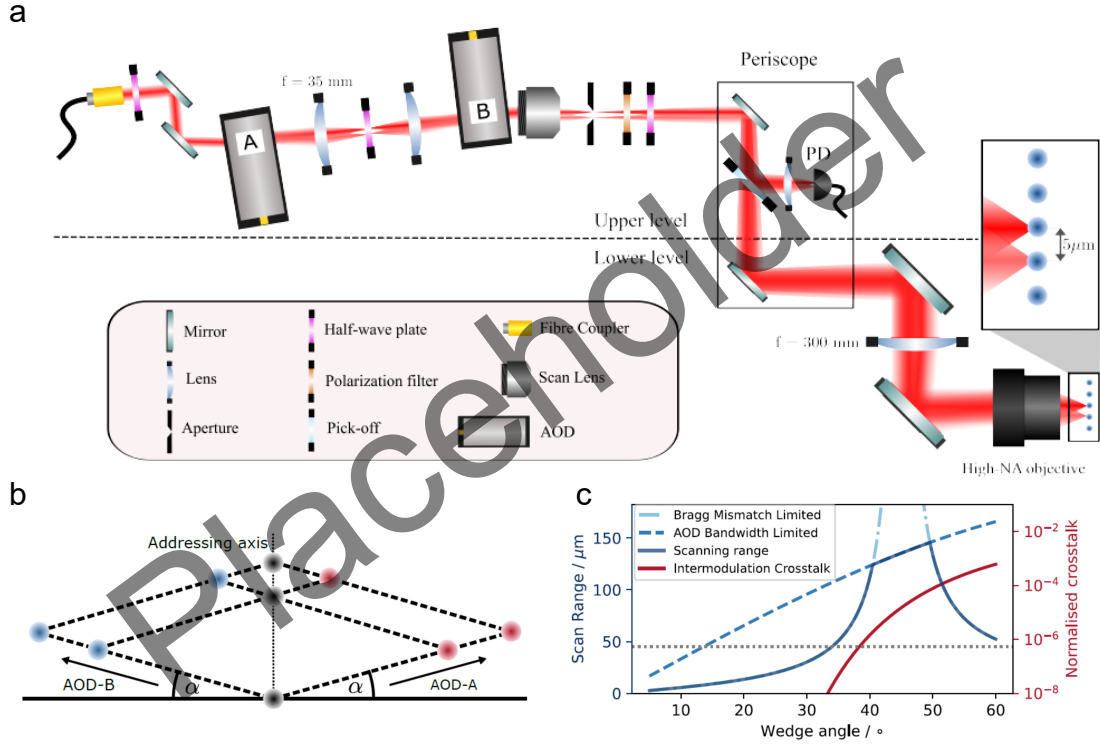


Figure 2.6: Asymptotic population decay for $k > 2$ qubits via numerical simulation.

This subsection is included as a brief pointer to a technique discussed only in outline in the manuscript SM (by analogy with *character RB* [24]). Expand into a full worked treatment if this state-engineering approach is pursued experimentally; otherwise keep as a short “future extensions” note and cross-reference from the Outlook (Section 2.6).

2.3.4 Scaling to $k > 2$ qubits

2.4 Analytic predictions for other noise types

Discuss ZZ-type errors, non-unitals, generalised depolarising channels, Stochastic vs coherent errors, etc.

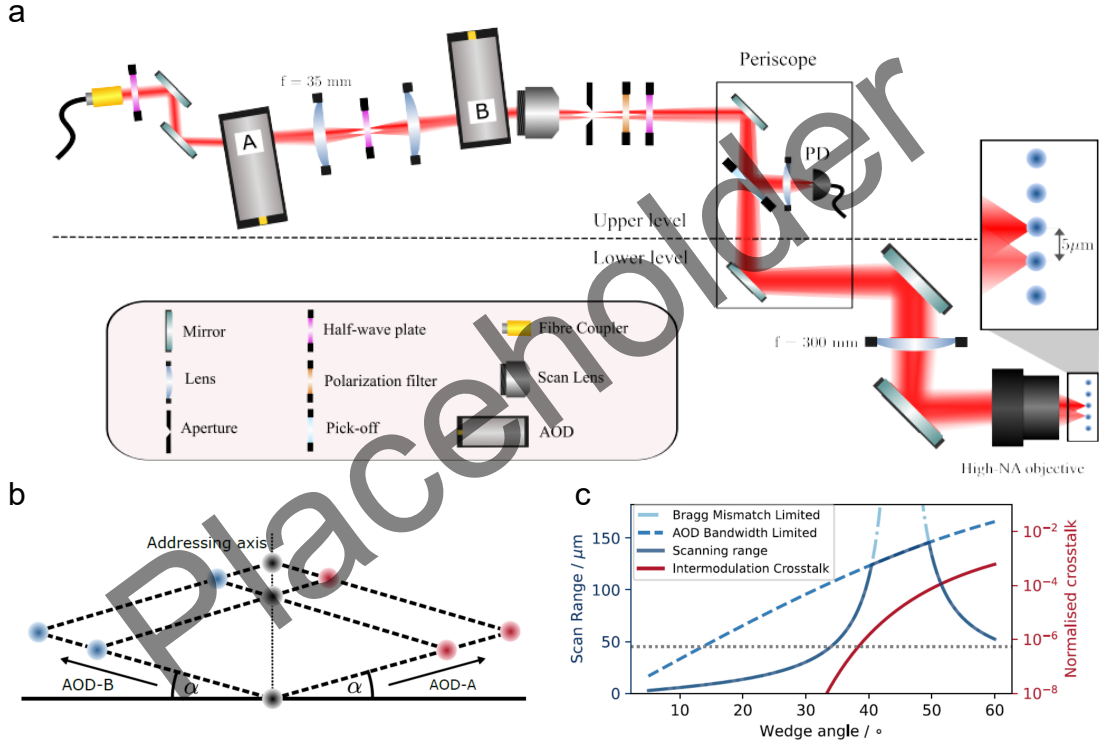


Figure 2.7: **Liouville-block structure for other noise types.**

2.5 Results

2.5.1 Discerning correlated and uncorrelated error

The 24 single-qubit Clifford gates are decomposed into native single-qubit rotations of the form $\hat{R}_x(\pi/2)$ and $\hat{R}_z(\pi/2)$: $\hat{\sigma}_x$ rotations are performed natively on resonance with the qubit transition, with relative phase set by modulating the 729 nm beam via an acousto-optic modulator (AOM), while $\hat{\sigma}_z$ rotations are performed virtually by updating the phase of subsequent $\hat{\sigma}_x$ rotations.

Z-basis measurements

k-copy RB is first demonstrated in a regime where common noise is the dominant error source, induced via a fixed 10 kHz frequency detuning of the 729 nm laser. Standard single-qubit RB decay curves (Section 1.6) are recorded alongside two-qubit correlated populations extracted via *k*-copy RB.

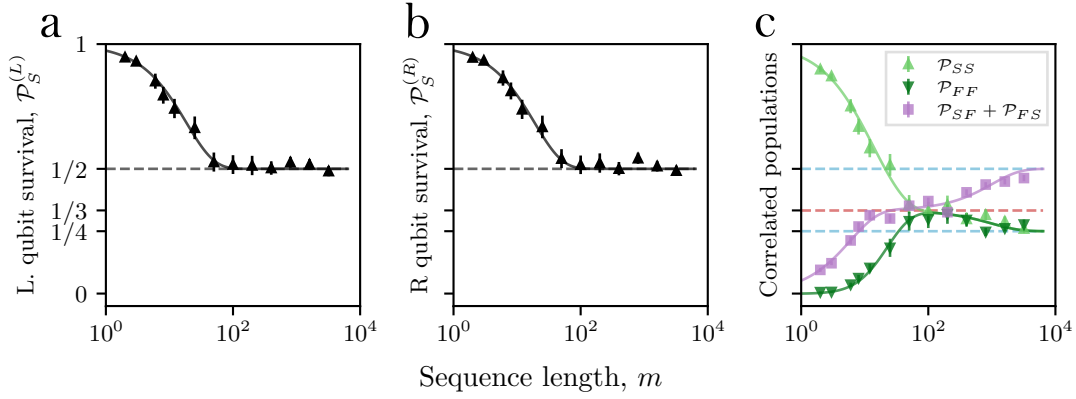


Figure 2.8: Exp data from fig 2 a,b,c

(a,b) single-qubit RB decays; (c) two-qubit correlated success population P_{SS} vs. sequence length; (d) parity contrast vs. sequence length, with example fringe insets. Correlated two-qubit probabilities discern correlated and uncorrelated error channels (adapted from manuscript Fig. 2). **Redraw for thesis; consider separating panels (a,b) [standard RB, Section 1.6 recap] from (c,d) [the k -copy-specific result] into two figures for pedagogical clarity.**

The correlated populations exhibit a clear intermediate asymptote at $m \approx 100$ Cliffords, corresponding to the initial state $|00\rangle$ approaching ρ_{sym} (Eq. (2.4)), followed by a slower decay to the fully depolarised two-qubit state by $m \approx 1000$. These two decay rates, corresponding to correlated and uncorrelated error respectively, are indistinguishable in the single-qubit decays and are only revealed through the correlated measurements made during k -copy RB. Fitting Eq. (2.1) gives extracted parameters $\varepsilon_{\text{corr}} = 0.051(3)$, $\varepsilon_L = 0.001(1)$, and $\varepsilon_R = 0.000(1)$.

Coherence measurements

A partial-tomography parity measurement (Section 2.3.2) is used to independently confirm occupation of the symmetric sector, quantifying overlap of the final state with $|\Psi^+\rangle$. The expected contrast asymptote $\mathcal{C} \rightarrow 1/3$ (Eq. (2.5)) is observed for $m \in [10, 100]$, corresponding to residual two-qubit coherence. Fitting Eq. (2.5) gives $\varepsilon_{\text{corr}} = 0.043(3)$ and $\varepsilon_L = \varepsilon_R = 0.0016(1)$; the increased depolarising rate

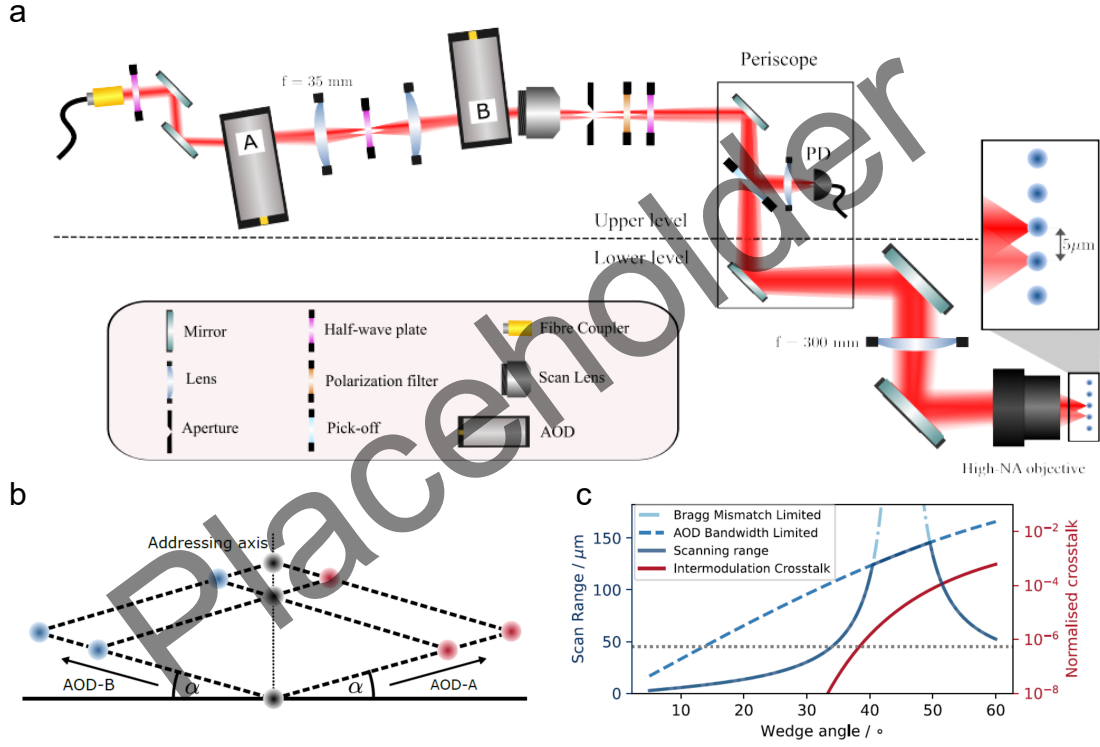


Figure 2.9: Exp data from fig 2 d + parity oscillations.

relative to the population fit is attributed to imperfect execution of the $R_{\pm\phi}(\pi/2)$ pulses used for the addressed differential rotation.

Add explicit discussion/reconciliation of the discrepancy between the $\varepsilon_L, \varepsilon_R$ values extracted from the population fit vs. the parity fit — the manuscript attributes this briefly to addressing-beam imperfections but a thesis chapter could quantify this more carefully (e.g. independently characterising the single-addressing system's contribution).

2.5.2 Controlled noise injection

Numerical simulations of correlated vs. uncorrelated noise

laser phase jitter

laser detuning scan

k-copy RB can isolate errors arising from shared control fields, making it useful both for device characterisation and for optimising global control parameters. This is

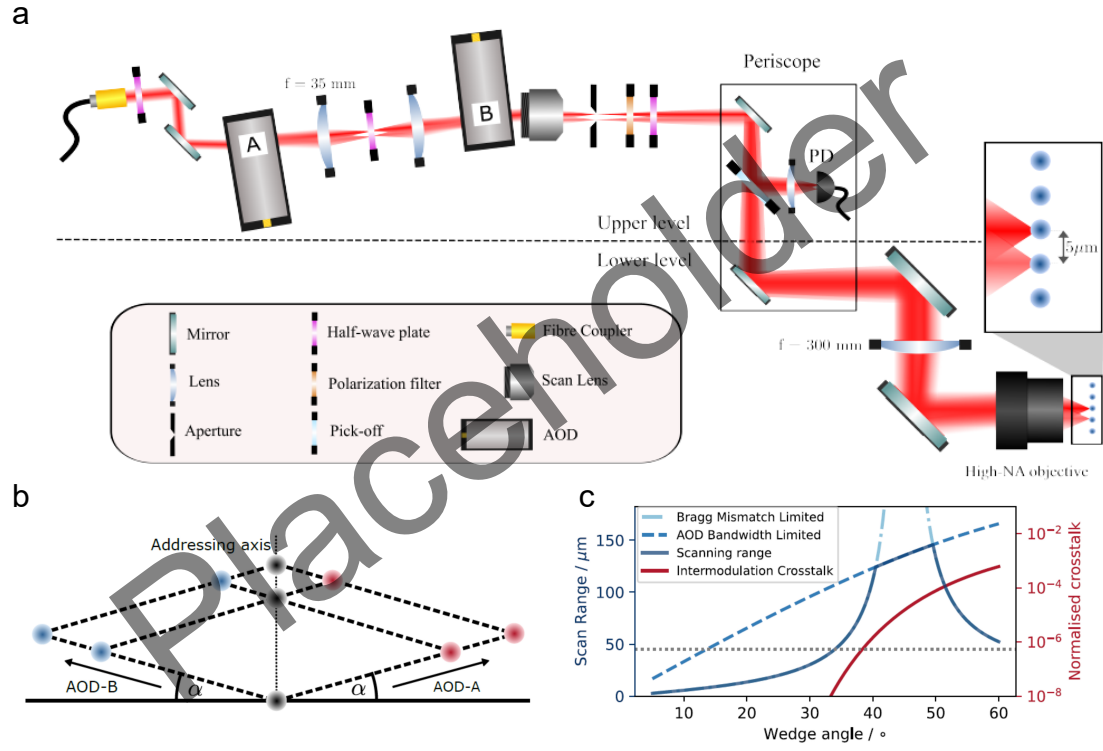


Figure 2.10: Numerical sim of param space and extracted fit uncertainties.

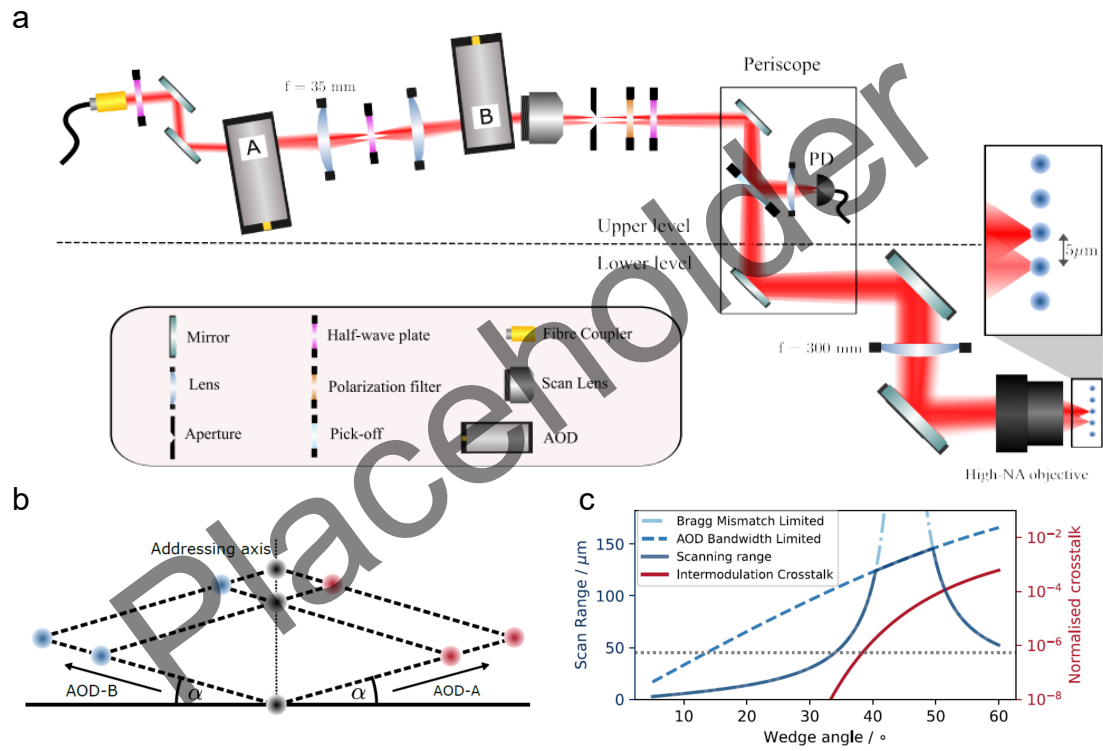


Figure 2.11: Injecting phase jitter calibration

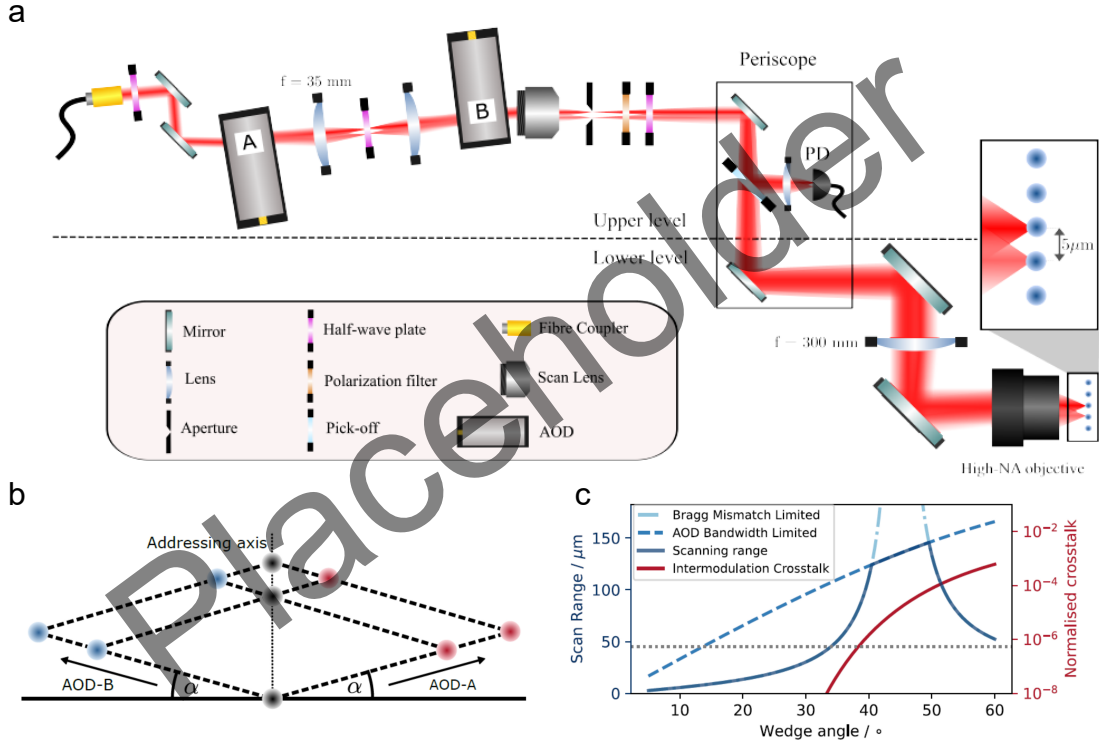


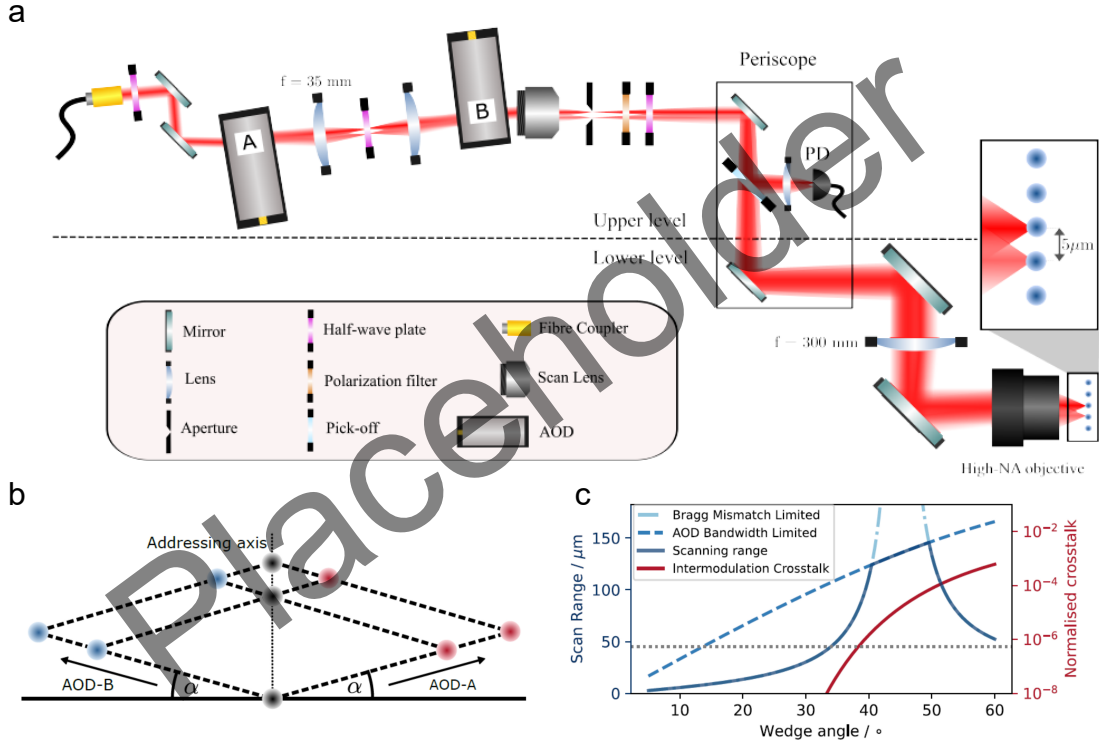
Figure 2.12: **figure 3 from paper.**

demonstrated by scanning the 729 nm laser frequency used for single-qubit rotations away from the bare qubit resonance.

Upper: example correlated populations for several detuning values. Lower: extracted correlated ($\varepsilon_{\text{corr}}$, red) and local ($\varepsilon_L, \varepsilon_R$, blue) error per Clifford vs. detuning, with quadratic fit. Correlated errors injected via detuning of a shared drive laser (adapted from manuscript Fig. 3). **Redraw for thesis.**

Each data point is obtained from a complete k -copy RB experiment with 11 sequence lengths up to $m = 1024$, using $r = 50$ random sequences and $s = 50$ shots per sequence, with correlated populations fit to Eq. (2.1) to extract $\{\varepsilon_{\text{corr}}, \varepsilon_L, \varepsilon_R\}$ at each detuning. The correlated error is observed to increase quadratically with detuning, while the local depolarising errors remain approximately constant, consistent with detuning of a shared drive producing common noise as defined in Section 1.8.2. This dependence is modelled as

$$\varepsilon_{\text{corr}} = \varepsilon_0 + \frac{\kappa}{3} \left(\frac{\delta_L - \delta_0}{\Omega} \right)^2, \quad (2.6)$$

Figure 2.13: **Calibrating stark shifts**

where ε_0 is the residual correlated error not originating from detuning, δ_0 accounts for the light-shifted resonance, $\Omega = 2\pi \times 100(1)$ kHz is the independently measured Rabi frequency, and κ depends on the Clifford decomposition to hardware-native gates. Fitting gives $\varepsilon_0 = 3.2(1) \times 10^{-3}$, $\kappa = 15.9(7)$, and $\delta_0 = 2\pi \times 0.80(5)$ kHz.

Derive Eq. (2.6) explicitly from the detuned-carrier propagator and average-fidelity argument given in the manuscript SM (“Noise injection methods”), i.e. $\hat{U}_\delta(t) = \exp(-\frac{i}{2}t(\Omega\hat{\sigma}_x - \delta\hat{\sigma}_z))$, the Haar-averaged fidelity $\mathcal{F}_{\text{avg}} = (|\text{Tr}(U_{id}^\dagger U_\delta)|^2 + 2)/6$, and the small- $\eta = \delta/\Omega$ expansion $\mathcal{F}_{\text{avg}} = 1 - \eta^2/3 + O(\eta^4)$. This was only sketched in the SM and deserves a full worked derivation here.

Single addressed stark shifts

2.5.3 Scaling to a multi-ion register

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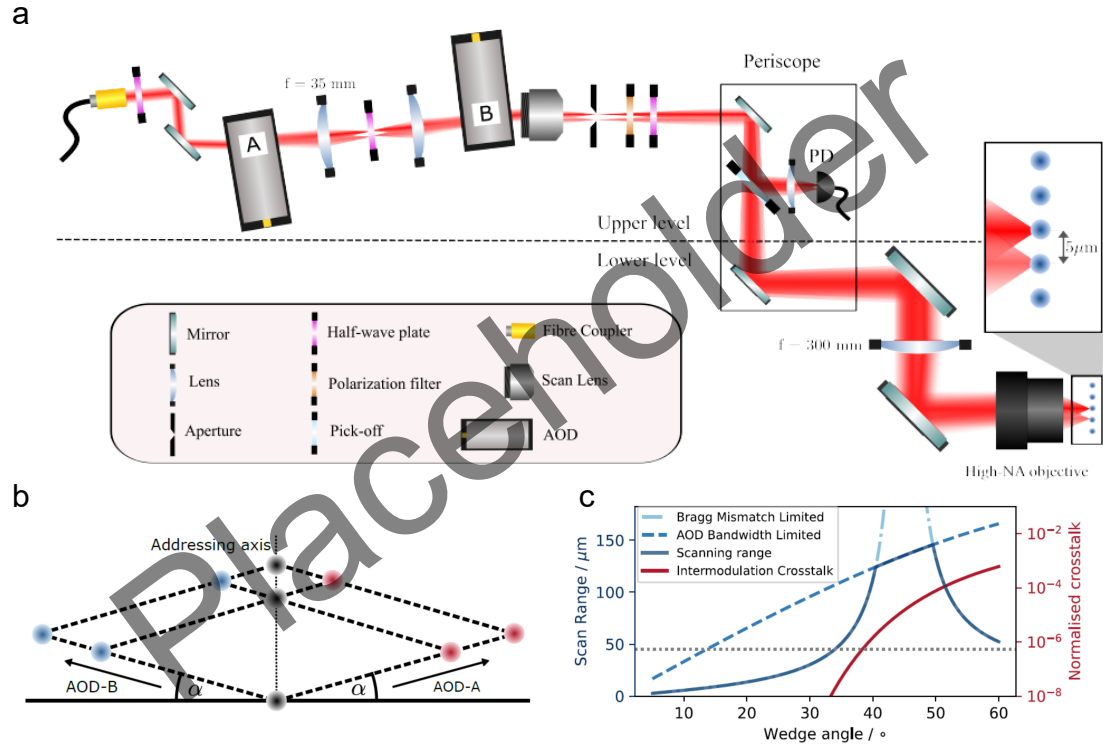


Figure 2.14: Stark shift supplemental figure.

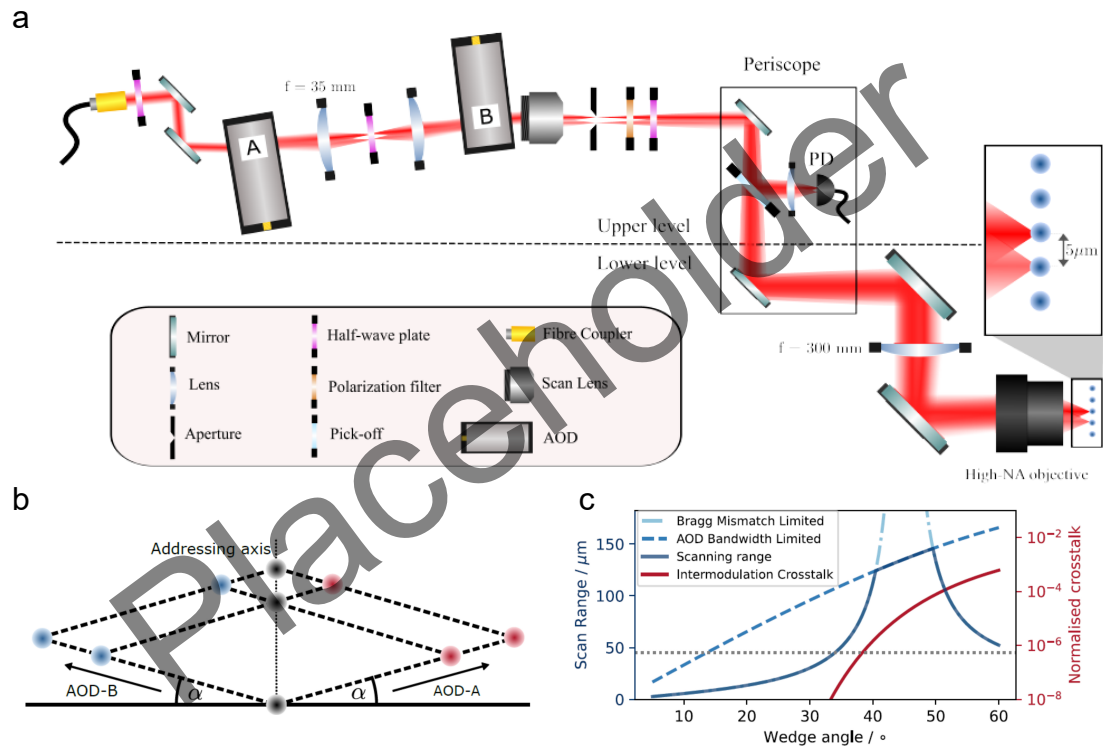
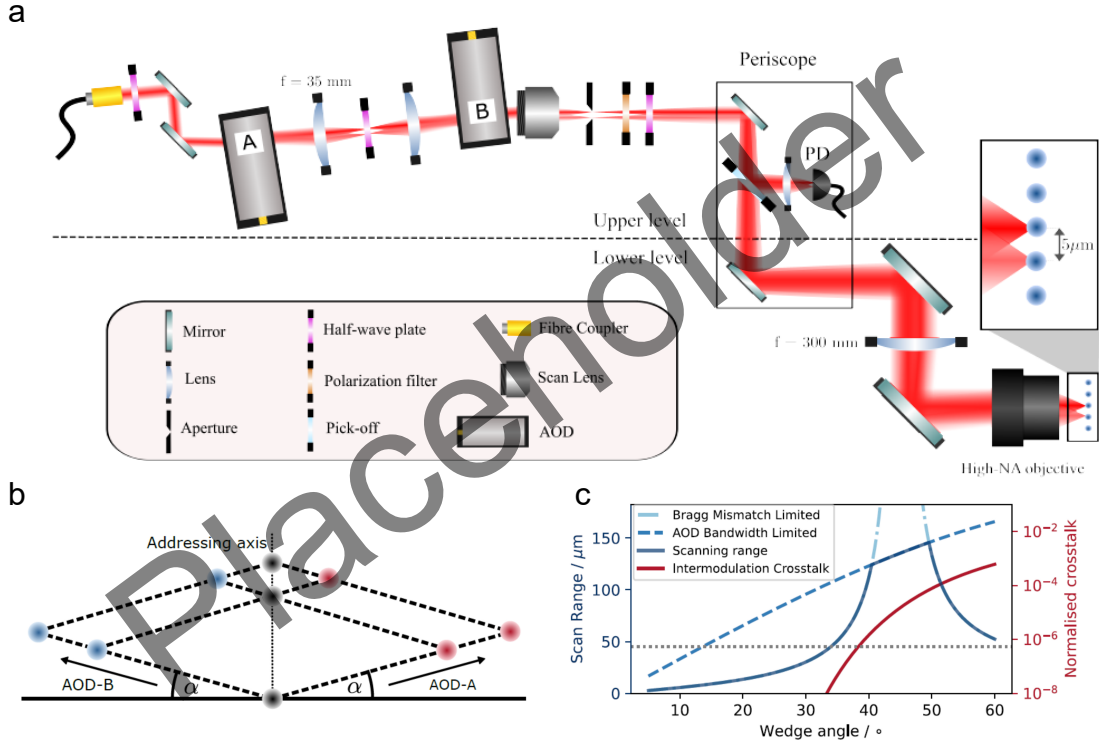


Figure 2.15: Calibrating global beam

Figure 2.16: **Figure 4 from paper.**

Pairwise k -copy RB extends simply to > 2 qubits: correlated populations are fit to Eq. (2.1) independently for each qubit pair. This provides a practical diagnostic of pairwise spatial correlations across a large register, though it does not uniquely determine higher-order (≥ 3 -body) correlated error terms (see Outlook, Section 2.6). For a register of k qubits, all $k(k-1)/2$ pairwise outcomes are extracted from the same shot record — no additional trials on the quantum register are required, though classical post-processing scales with the number of pair combinations. The protocol itself requires only synchronised application of identical single-qubit Clifford sequences and spatially resolved readout.

(a) Extracted pairwise correlated error $\varepsilon_{\text{corr}}^{(i,j)}$ for all ion pairs in an 8-ion chain. (b) Sum of local depolarising errors $(\varepsilon_L + \varepsilon_R)^{(i,j)}$. (c) Ratio of excess correlated noise to local errors. Correlated and local error per Clifford measured for all qubit pairs in an 8-ion chain (adapted from manuscript Fig. 4). **Redraw for thesis.**

This scaling is demonstrated on an eight-ion chain, with parallel single-qubit rotations driven by an elliptical 729 nm beam of major-axis waist radius $52(3) \mu\text{m}$ aligned to the ion chain. Addressing multiple qubits in parallel with a single shared drive is common to both ion-trap and neutral-atom platforms owing to its experimental simplicity, but unequal illumination across the register leads to position-dependent over- and under-rotation errors. With axial confinement of 0.5 MHz, the chain extent is $35 \mu\text{m}$, giving an expected Rabi-frequency difference of $11(1)\%$ between the outer and inner ions (single-qubit gates were optimised for the inner ions, $i = 4$). The extracted pairwise correlations show the resulting spatially dependent error structure: both neighbouring and distant outer-ion pairs exhibit large correlated errors, and in particular the maximally separated ions $i = 1$ and $j = 8$ experience similar under-rotations, producing a strong correlated error despite their spatial separation.

To improve sensitivity to both correlated and local contributions simultaneously, a common error offset was intentionally added by randomising the phase of each implemented $R_x(\pi/2)$ pulse (drawn from a normal distribution with standard deviation $s_\phi = 0.19 \text{ rad}$, producing $\varepsilon_{\text{corr}}^{(0)} = 0.025$). This separates the relevant exponential decay rates and improves the stability of the correlated-population fits.

Add the noise-injection methodology in more detail here, including the complementary Stark-shift-beam injection experiments from the manuscript SM (App. “Noise injection methods”, Fig. S2): two addressing configurations (correlated two-spot vs. independent single-spot Stark shifts) used to independently calibrate and validate the $\varepsilon_{\text{corr}}$ vs. $\varepsilon_L, \varepsilon_R$ separation against a known injected error rate.

2.6 Conclusions and outlook

k-copy RB efficiently discerns correlated and uncorrelated error channels acting on a multi-qubit system. Because of the symmetry enforced by twirling over the diagonal Clifford representation $Cl^{\otimes k}$ (Section 1.7.2), structured errors drive the decay of correlated populations into invariant subspaces in a way that is directly

observable without full process reconstruction. Given a specified error model, these correlated decays can be used to directly quantify the magnitude of the individual error channels contributing to the total error per Clifford.

This method has been proposed and experimentally demonstrated, and its application extended to an eight-qubit register through pairwise comparisons (Section 2.5.3). This approach permits highly efficient parallel data collection: the time required on the device is independent of qubit number, and the protocol requires exclusively local operations followed by classical post-processing of correlated survival rates.

Expand this Outlook section substantially for the thesis, drawing together the forward-looking threads seeded earlier in this chapter and in Chapter ??:

- **Extension to genuine ≥ 3 -body correlated error terms, beyond the pairwise diagnostic of Section 2.5.3.**
- **Experimental characterisation of ZZ -type entangling errors (Section 1.8), not demonstrated here.**
- **Sample-cost / Fisher-information analysis for multi-exponential fitting — flagged as an open numerics question in the manuscript SM and not yet resolved.**
- **Non-Markovian (temporally correlated) noise, via repeated identical Clifford RB sequences applied at different times on a single qubit, exploiting the same correlated-outcome analysis used here for spatial correlations.**
- **Character-RB-style initial-state engineering (Section 2.3) to isolate single exponential decays and reduce sample cost.**

Appendices

A

Appendix

A.1 Generating Ions

A.2 Extracting Laser Offset and Magnetic Field

A.3 Experimental Control

A.4 Creating Squeezed States

A.5 Detuning errors

To exemplify the method for calculating the effect of an error channel on a single qubit gate, we consider the simple case of a coherent detuning error. We define the Fidelity of a quantum channel $\mathcal{E}$ with respect to a unitary U as

$$F(\mathcal{E}, U) = \int d\psi \langle \psi | U^\dagger \mathcal{E}(|\psi\rangle \langle \psi|) U | \psi \rangle, \quad (\text{A.1})$$

where the integral is taken over the uniform Haar measure on the space of pure states. We shall see that the language used for this analysis of single qubit gate fidelities closely resembles the group theory approach describing randomised benchmarking. The Haar measure is relevant here to consider the fidelity of the operation independent of the initial state (assuming it is pure), which is exactly the enforced criteria through Clifford twirling during RB. For a single qubit, this

can be expressed as an integral over the Bloch sphere. If we further assume that the error channel is unitary, as is the case of a detuned channel considered here, we shall find that the fidelity follows

$$F(V, U) = \frac{1}{d(d+1)} \left(|\text{Tr}(W)|^2 + d \right), \quad (\text{A.2})$$

where V is the error unitary, $W = U^\dagger V$ and d is the dimension of the Hilbert space. Here we shall give a derivation of this result, by considering the symmetry imposed by the Haar measure. The standard fidelity of two unitaries following from equation A.1 is given by

$$\begin{aligned} F(V, U) &= \int d\psi \langle \psi | U^\dagger V | \psi \rangle \langle \psi | V^\dagger U | \psi \rangle, \\ &= \int d\psi |\langle \psi | W | \psi \rangle|^2. \end{aligned} \quad (\text{A.3})$$

Using the identity $\text{Tr}(A)\text{Tr}(B) = \text{Tr}(A \otimes B)$, we can rewrite the fidelity as a two-copy expectation value,

$$F(V, U) = \text{Tr} \left((W \otimes W^\dagger) \int d\psi (|\psi\rangle \langle \psi|)^{\otimes 2} \right), \quad (\text{A.4})$$

where we have now separated the fidelity into a part dependent on the error channel, and a part equivalent to the second moment of the Haar measure, explicitly rewritten as

$$\mathcal{M}_2 = \int d\psi (|\psi\rangle \langle \psi|)^{\otimes 2}. \quad (\text{A.5})$$

We now consider two apparent symmetries of $\mathcal{M}_2$: first, the Haar measure is invariant under unitary transformations, and so our expression must commute with $U \otimes U$ for any unitary U ; second, the two states in the tensor product are identical, and so $\mathcal{M}_2$ must be invariant under the swap operator S which exchanges the two copies. These two symmetries are sufficient to show that $\mathcal{M}_2$ is proportional to the projector onto the symmetric subspace of the two-copy Hilbert space, which is given by

$$\Pi_{\text{sym}} = \frac{1}{2}(\mathbb{I} + S). \quad (\text{A.6})$$

To normalise the moment (as the Haar measure is a probability distribution), we enforce $\text{Tr}(\mathcal{M}_2) = 1$. In this two-copy expression, it is apparant that $\text{Tr}(I) = d^2$ and $\text{Tr}(S) = d$, resulting in,

$$\mathcal{M}_2 = \frac{1}{d(d+1)} (\mathbb{I} + S). \quad (\text{A.7})$$

Incorporating this result into the fidelity expression, we find

$$F(V, U) = \frac{1}{d(d+1)} \text{Tr} \left[(W \otimes W^\dagger)(\mathbb{I} + S) \right], \quad (\text{A.8})$$

where we can evaluate the two terms using the identities $\text{Tr}(A \otimes B) = \text{Tr}(A)\text{Tr}(B)$ and $\text{Tr}((A \otimes B)S) = \text{Tr}(AB)$, the latter of which is highly relevant to the mechanism of virtual distillation, described in chapter ???. This results in the final expression for the fidelity of a unitary error channel with respect to a target unitary, given above by equation A.2.

Considering the explicit case of a $R_x(\pi/2)$ gate with a coherent detuning error, we can write the error unitary as

$$V = \exp \left(-i \frac{\pi}{4} \left(\hat{\sigma}_x + \frac{\delta}{\Omega} \hat{\sigma}_z \right) \right), \quad (\text{A.9})$$

where δ is the detuning from resonance, and Ω is the Rabi frequency. The ideal unitary is given by $U = \exp(-i \frac{\pi}{4} \hat{\sigma}_x)$, yielding trace of the product $W = U^\dagger V$ as

$$\text{Tr}(W) = 2 - \frac{1}{2} \left(\frac{\delta}{\Omega} \right)^2 + \mathcal{O} \left(\frac{\delta}{\Omega} \right)^4, \quad (\text{A.10})$$

where we have assumed the detuning is small compared to the Rabi frequency. Inserting this into equation A.2, we find the fidelity of the detuned gate is given by

$$F(V, U) = 1 - \frac{1}{3} \left(\frac{\delta}{\Omega} \right)^2 + \mathcal{O} \left(\frac{\delta}{\Omega} \right)^4. \quad (\text{A.11})$$

To convert this to an error contribution to a single qubit Clifford gate, we introduce the scalar κ , which depends on the specific decomposition of the Clifford gate set into the native gate set of the system. The error contribution is then given by

$$\epsilon = \kappa(1 - F(V, U)) = \frac{\kappa}{3} \left(\frac{\delta}{\Omega} \right)^2 + \mathcal{O} \left(\frac{\delta}{\Omega} \right)^4. \quad (\text{A.12})$$

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